A special biserial algebra with finite projective dimension three

Submission for TLMC problem 00000000551

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Claim and scope. The conjecture states the universal bound $\operatorname{findim}\Lambda \leq 2$ for special biserial algebras. We give a finite-dimensional complex special biserial algebra A with a finitely generated module S_3 of projective dimension exactly 3. Consequently $\operatorname{findim} A \geq 3$. The reference in the source to computations for cyclic string algebras does not restrict its stated universal quantifier; our example is acyclic. We do not claim a classification or an upper bound for all special biserial algebras.

The algebra and its actual path presentation. Let Q be the linearly oriented quiver

$$3 \longrightarrow 2 \longrightarrow 1 \longrightarrow 0,$$

and let I kill every path of length at least two. Set $A = \mathbb{C}Q/I$. Explicitly, $A = \mathbb{C}^4 \oplus \mathbb{C}^3$, with

$$(x, m)(y, n) = \left((x_i y_i)_{i=0}^{3}, (x_i n_i + m_i y_{i+1})_{i=0}^{2}\right).$$

This is the standard square-zero extension with distinct left and right diagonal actions, implemented by Mathlib's `TrivSqZeroExt`.

The formal proof also checks the path presentation, rather than presuming it. In the incidence algebra T of the four-element chain, the interval basis E_{ij}, $0 \leq i \leq j \leq 3$, satisfies $E_{ij}E_{kl} = \delta_{jk}E_{il}$ whenever the indices concatenate. There is exactly one actual quiver path from j to i when $i \leq j$, and none otherwise; its length is $j - i$. The formal basis and the multiplication law therefore identify T with $\mathbb{C}Q$. The surjective algebra homomorphism

$$\phi : T \longrightarrow A, \qquad f \longmapsto ((f(i,i))_{i=0}^{3}, (f(i,i+1))_{i=0}^{2})$$

has kernel

$$I = \operatorname{span}_{\mathbb{C}}\{E_{02}, E_{13}, E_{03}\} = \langle E_{02}, E_{13}, E_{03}\rangle.$$

The proof constructs the resulting quotient algebra equivalence. Thus the relations are genuine paths of length at least two, and no nonzero vertex or arrow relation has been introduced.

Every vertex has at most one incoming and one outgoing arrow. Every product of two arrows lies in I, so each arrow has zero permitted predecessors and successors modulo I. All paths of length at least two belong to I. These verify the standard special biserial conditions; the algebra is also a string algebra. The definitions used are those of Erdmann–Holm–Iyama–Schroer [1, pp. 168–170]; their definition of the little finitistic dimension is the supremum over finitely generated modules of finite projective dimension [1, p. 160].

Concrete projective modules and exact sequences. Write $S_i = \mathbb{C}$, with $(x, m)z = x_i z$. These are actual left A-modules. Each is finitely generated, since the diagonal coordinate map $A \to S_i$ is surjective. The module S_0 is a direct summand of the regular module A, using the idempotent at vertex 0, and hence is projective.

For $1 \leq i \leq 3$, let $P_i = \mathbb{C}^2$ with action

$$(x, m)(u, v) = (x_i u,\ m_{i-1}u + x_{i-1}v).$$

The maps $A \to P_i$, $(x, m) \mapsto (x_i, m_{i-1})$, and $P_i \to A$, $(u, v) \mapsto (ue_i, v\alpha_{i-1})$, are A-linear and compose to the identity on P_i. Thus each P_i is a direct summand of the free regular module, and is projective. The coordinate maps give short exact sequences

$$0 \longrightarrow S_{i-1} \xrightarrow{v \mapsto (0,v)} P_i \xrightarrow{(u,v) \mapsto u} S_i \longrightarrow 0 \qquad (1 \le i \le 3).$$

Injectivity, surjectivity and equality of image and kernel are checked on the actual module maps.

The first sequence cannot split. If an A-linear section $s : S_1 \to P_1$ existed, then $s(1) = (1, c)$ for some $c \in \mathbb{C}$. The arrow $\alpha_0 : 1 \to 0$ acts by zero on S_1, but

$$\alpha_0 s(1) = (0, 1) \neq 0 = s(\alpha_0 \cdot 1),$$

a contradiction. Hence S_1 is not projective.

Projective dimension and the contradiction. The three short exact sequences with projective middle terms imply $\operatorname{pd} S_3 \le 3$. Dimension shifting also gives

$$\operatorname{pd} S_3 < 3 \implies \operatorname{pd} S_2 < 2 \implies \operatorname{pd} S_1 < 1,$$

which would make S_1 projective. Therefore $\operatorname{pd} S_3 = 3$. The Lean proof uses Mathlib's `CategoryTheory.projectiveDimension`, defined through vanishing of the actual Ext groups, and its short exact sequence dimension-shifting theorem. No dimension is assigned by definition.

The formal `finitisticDimension` is the genuine supremum of those projective dimensions for all finitely generated objects of `ModuleCat A` having finite projective dimension. Its indexing set contains the explicitly constructed S_3, so the supremum is at least 3. The library's bottom value for the zero object has no effect on this positive lower bound. This disproves the stated universal bound 2.

Formal files. `lean/Main.lean` contains the full construction and proof, including the path basis, quotient equivalence and special biserial certificate. Its conclusions include

`simple_three_projectiveDimension`, `counterexample`.

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

Verification commands, all-namespace axiom audit, fresh-kernel receipts and file hashes accompany the submission. The complete original bilingual problem is preserved in `SOURCE.md`.

References

[1] K. Erdmann, T. Holm, O. Iyama and J. Schroer, *Radical embeddings and representation dimension*, Advances in Mathematics **185** (2004), 159–177. doi:10.1016/S0001-8708(03)00169-5.